The Bi Cluster Test at $n = 4$: Partial No, Structural Yes

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Abstract

We test whether the principal minors $d_k(n) = \det M^{(k)}[D_n]$ of the staircase Hecke element Π_q sit inside Bi's cluster algebra on the Hernandez–Leclerc subcategory $\mathcal{C}_{v,\beta}$. At $n = 4$ the direct identification $d_k \leftrightarrow D_k$ fails on three counts: count, exchange form, and ambient ring. The failure is diagnostic: the staircase case $v = \delta(\beta)$ is precisely the degenerate case of Bi's framework, where the new content collapses to Geiß–Leclerc–Schröer. Structural matches in rank and generator count suggest the live alternative is Sandvik's monodromic-Hecke / Soergel-bimodule route.

1 Setup — the question

Bi (arXiv:2602.11559) constructs, for each reduced word β of $w_0 \in W$ and each $v \leq \delta(\beta)$ in the higher Bruhat order, a monoidal subcategory $\mathcal{C}_{v,\beta}$ of the Hernandez–Leclerc category whose Grothendieck ring $K_0(\mathcal{C}_{v,\beta})$ carries a cluster algebra structure. The mutable cluster variables are (quantum) minors of Bott–Samelson type; frozen variables are boundary Plücker coordinates.

Our natural test case: the staircase reduced word

$$w_0 \;=\; s_1 s_2 s_3 \cdot s_1 s_2 \cdot s_1 \qquad \text{in } S_4 \;=\; W(A_3),$$

with length $\ell(w_0) = 6 = |D_4|$ (the descent-paired lattice size at $n = 4$). The *staircase Hecke element* is

$$\Pi_q \;=\; (1+T_1)(1+T_2)(1+T_1)(1+T_3)(1+T_2)(1+T_1) \;\in\; H_q(S_4).$$

It is the Bott–Samelson element at this reduced word. Its principal minors on the descent-paired set $D_n \subset S_n$ have been the central object of our April 2026 work; the determinantal identity

$$d_{2j+1}(n) \cdot d_{2j-1}(n) \;=\; q^{|D_n|}\, d_{2j}(n)^2 \qquad \text{(Hirota-type)}$$

has been proved at $j = 1$ and $j = 2$ and heavily supported at $j = 3$.

We asked two sharp questions:

H1. Are the $d_k(n)$ individual cluster variables in Bi's seed on the staircase?

H2. Is the Hirota identity $d_{2j+1} d_{2j-1} = q^{|D_n|} d_{2j}^2$ an exchange relation?

A "yes" would identify Π_q's integrable structure with the KKKO/Bi cluster structure on $K_0(\mathcal{C}_{v,\beta})$; it would be a categorical origin for the palindromic factor.

2 Data at $n = 4$

Direct Hecke computation (see Appendix A) gives

$$d_1(4) \;=\; q^6, \tag{1}$$

$$d_2(4) \;=\; q^9(q+1)^3(2q+1), \tag{2}$$

$$d_3(4) \;=\; q^{18}(q+1)^6(2q+1)^2 \;=\; d_2(4)^2. \tag{3}$$

The first q-shifted pair identity checks:

$$d_3(4) \cdot d_1(4) \;=\; q^{18}(q+1)^6(2q+1)^2 \cdot q^6 \;=\; q^6 \cdot q^{18}(q+1)^6(2q+1)^2 \;=\; q^6 \cdot d_2(4)^2,$$

with shift exponent $|D_4| = 6$, consistent with the general Hirota conjecture.

So at $n = 4$ there are only *three* distinct d_k quantities, and the middle one has a clean factorisation into three "staircase atoms" q, $(q+1)$, $(2q+1)$.

3 Bi's initial seed for the staircase $w_0 = (1, 2, 3, 1, 2, 1)$ in A_3

From Bi §2 and Theorem 3.1, the initial seed for $\mathcal{C}_{w_0,\beta}$ at the staircase word has six vertices, with the standard frozen choice $\{3, 5, 6\}$ (boundary Plückers) and mutable $\{1, 2, 4\}$. The cluster variables are the six positive-root generalised minors of $\mathbb{C}[N_-]$ for SL_4, matching Geiß–Leclerc–Schröer:

$$D_1 \;=\; \Delta_{2,1} \tag{4}$$
$$D_2 \;=\; \Delta_{23,12} \tag{5}$$
$$D_3 \;=\; \Delta_{234,123} \qquad \text{(frozen, size 3)} \tag{6}$$
$$D_4 \;=\; \Delta_{3,2} \tag{7}$$
$$D_5 \;=\; \Delta_{34,23} \qquad \text{(frozen, size 2)} \tag{8}$$
$$D_6 \;=\; \Delta_{4,3} \qquad \text{(frozen, size 1)} \tag{9}$$

By Plücker size: three size-1 minors, two size-2 minors, one size-3 minor. This is the A_3 positive-root system organised by height.

The exchange relation (Bi eq. 4.16) is the standard three-term Fomin–Zelevinsky form:

$$D_{k-1} \cdot \mu_{k-1}(D_{k-1}) \;=\; q^A\, D_k \cdot D_{(k-1)^-} \;+\; q^B\, D_{k+1} \cdot D_{k^-}.$$

Remark 1. For $v = w_0 = \delta(\beta)$ — the full staircase — Bi's new content collapses back to the classical GLS seed on $\mathbb{C}[N_-]$ (Theorem 3.11, final sentence). In Bi's framework, the staircase $v = w_0$ is the *degenerate / boundary* case. Bi's innovation operates for $v < \delta(\beta)$, where the seed is genuinely smaller than the full GLS seed and the cluster structure detects sub-Bruhat-order information.

4 Three mismatches

Observation 2 (Count)**.** Bi's seed at $v = w_0$ in A_3 has **6** cluster variables. We have only **3** distinct d_k quantities at $n = 4$. If d_k were each an individual cluster variable, we would need three variables out of six. No natural assignment presents itself: the d_k have no canonical subset structure inside the positive-root system.

Observation 3 (Exchange form)**.** Bi's exchange is three-term (two monomials on the right). Our Hirota identity is two-term bilinear:

$$d_{2j+1}\, d_{2j-1} \;=\; q^{|D_n|}\, d_{2j}^2.$$

A two-term bilinear identity cannot arise from a single mutation of a three-term exchange unless one of Bi's monomials vanishes identically — which does not happen in a generic cluster algebra. Even allowing aggregation into products of several D_k, the right-hand side of a cluster exchange never collapses to a single monomial in the generic case.

Observation 4 (Ambient ring)**.** Bi's D_k live in the quantum unipotent coordinate ring $\mathbb{C}_q[N_-]$, a quotient of $U_q(\mathfrak{n}_-)$. Our d_k live in the Hecke algebra $H_q(S_n)$. These are related at the dual level by Schur–Weyl duality

$$H_q(S_n) \curvearrowright V^{\otimes n} \curvearrowleft U_q(\mathfrak{sl}_n),$$

but the pairing between Bott–Samelson Hecke elements and Plücker-type minors is nontrivial. The natural image of a Bott–Samelson element of H_q in $\mathbb{C}_q[N_-]$ — e.g. via the Kashiwara–Lusztig global basis — is typically a *product* of several generalised minors, not a single one.

These three observations together kill H1 and H2 in the literal reading $d_k \leftrightarrow D_k$.

5 Structural positives (to guide next experiment)

The failure has a pattern, and the pattern is informative.

Observation 5 (Rank-count match)**.** $|D_4| = 6$ equals the number of Bi cluster variables, which equals $\ell(w_0) = |\Phi^+(A_3)|$. The descent-paired lattice size equals the Bi seed size equals the number of positive roots. Both sides count the generators of the Bott–Samelson / positive-root system, so this is not coincidence — but it is the *outer* count matching, not the individual generators.

Observation 6 (Size distribution)**.** Bi's six generators split by Plücker size as $(3, 2, 1)$: three size-1, two size-2, one size-3. Our d_k split by level as $(k = 1, k = 2, k = 3)$, i.e. three levels with one quantity each. Both are "staircase" sequences, but with opposite monotonicity (Bi's is decreasing, ours is uniform-by-level). The natural question: is there a bijection pairing d_k with the aggregated monomial over size-$(n-k)$ minors? Numerically this would mean

$$d_k(4) \stackrel{?}{=} \prod_{\text{size-}(n-k)\text{ minors } D_i} D_i^{a_i}$$

for some exponents a_i, evaluated at the GLS parametrisation. This is testable but was not carried out here.

Observation 7 (Bott–Samelson interpretation)**.** Π_q is precisely the Bott–Samelson element of H_q at the reduced word $(1, 2, 1, 3, 2, 1)$. In the Soergel-bimodule category this is $BS(w_0)$, whose class in $K_0(\mathrm{SBim}) \cong H_q$ is the product of Kazhdan–Lusztig generators C'_{s_i}. This is the **Sandvik / monodromic Hecke** route (arXiv:2604.15582), not the Bi cluster route.

The two categorical survivors from our three-routes analysis operate at different levels of the geometric Langlands tower:

- **Bi:** quantum-group reps, K_0 of a monoidal subcategory of finite-dim $U_q(\widehat{\mathfrak{sl}_n})$-modules.
- **Sandvik:** Soergel bimodules / parity sheaves on G/B, K_0 of Hecke-module categories.

Π_q is a Hecke-algebra element *on the nose*, so the Soergel / parity-sheaf side is the direct home.

6 Conclusion

Both H1 and H2 are **false** at $n = 4$ under the direct identification $d_k \leftrightarrow D_k$. The numerical count (6 vs 3), the exchange form (three-term vs two-term), and the ambient ring ($\mathbb{C}_q[N_-]$ vs $H_q(S_n)$) each independently rule out the naive correspondence.

The reason is structural: the staircase case $v = \delta(\beta)$ is precisely the *degenerate* case of Bi's framework, where the seed reduces to the classical GLS seed and Bi's new content — the $v < \delta(\beta)$ refinement — is absent. We chose the test case that sits at the boundary of Bi's theorem, which is exactly where it has least to say.

The negative answer is diagnostic. Candidates remaining:

(a) d_k as a *product / aggregation* of Bi cluster variables at a fixed Plücker size — requires computing Bi classes in $\mathbb{Z}[q]$ and matching the $(q+1)^3(2q+1)$ factorisation of $d_2(4)$ against products of GLS minors.

(b) d_k as a Grothendieck class in a different monoidal category — **Sandvik monodromic Hecke**, where $BS(w_0)$ is a parity sheaf and self-duality gives the palindromic factor a priori. This is the structurally natural home.

(c) The Hirota identity as the *shadow* of a higher-level identity in Bi's framework (double mutation, interval, product seed) at $v < \delta(\beta)$, projected back to the staircase by specialisation. More speculative.

The cleanest next experiment is (b): compute Soergel-bimodule classes for $BS(w_0)$ and check whether the Hirota identity is a shadow of a Grothendieck-group relation in $K_0(\mathrm{SBim})$. This is consistent with the *KL basis unification* theme of the three April 2026 papers, and it is the route where Π_q appears on the nose rather than after a Schur–Weyl transposition.

A Scripts and data

- `/home/clio/projects/scratch/2026-04-21-staircase-n4-bi-data.py` — direct Hecke computation of $d_1(4), d_2(4), d_3(4)$ and the first-pair identity check.
- `/home/clio/projects/scratch/2026-04-21-staircase-n4-data.md` — tabulation of the staircase data at $n = 4$ in human-readable form.
- `/home/clio/projects/scratch/2026-04-21-bi-cluster-extract.md` — extract of Bi §2–3 with the initial-seed specialisation to $v = w_0$ in A_3.